FROM API CALLS TO AGENTS

FIRST PRINCIPLES

2025-2026: THE YEAR OF AI AGENTS

The landscape of software development has shifted dramatically with the emergence of AI agents. Tools like Claude Code and Codex operate directly within terminals or cloud sandboxes, leading to significant productivity gains by automating file edits, testing, and debugging.

TOOL	PRIMARY ENVIRONMENT	KEY FEATURES
Claude Code	Local Terminal	Reads codebase, edits files, runs tests, commits code.

AUTOCOMPLETE → CHAT → AGENT

2021

AUTOCOMPLETE

GitHub Copilot

AI suggests the next line of code within your editor. The human developer remains the primary driver of logic and structure.

2023

CHAT ASSISTANT

ChatGPT / Claude

Conversational interface where users describe problems. AI provides code blocks that the human manually copy-pastes into projects.

2025

AGENT

Claude Code / Codex

Goal-driven autonomy. The agent plans, executes steps, uses tools, and iterates until the objective is achieved.

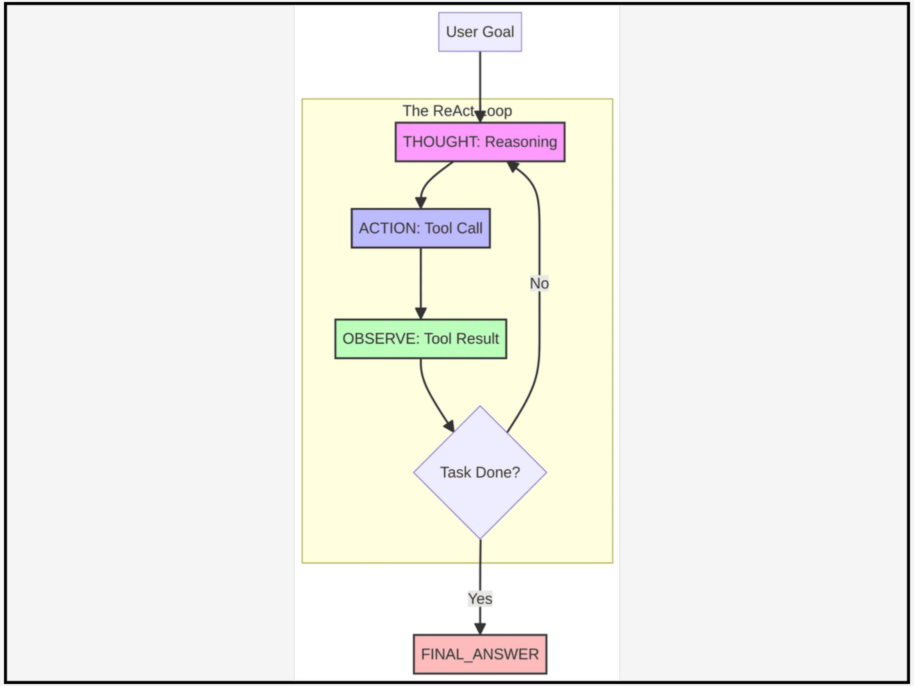
DEFINITION

LLM + Tools + Loop

LLM: The reasoning engine that understands intent and plans steps.

Tools: External functions the LLM can call to interact with the world.

Loop: The iterative process that continues until the goal is met.



THE REACT PATTERN: REASONING + ACTING

WHY THE LOOP OUTPERFORMS ONE-SHOT

ITERATIVE FEEDBACK

The agent loop acts as a powerful extension of Chain-of-Thought reasoning. Every iteration represents a new forward pass that incorporates real-world data from tools, allowing the model to observe results and adjust its strategy.

REAL-WORLD FEEDBACK

Tools provide external data that isn't present in the static model weights. The agent interacts with the environment to gain fresh context.

ITERATIVE CORRECTION

Agents can try a solution, see it fail, and adjust their strategy—just like a human developer debugging code in real-time.

PERSISTENCE

This architecture allows tools to work autonomously for hours on complex, multi-file features that would overwhelm a single prompt.

DEMYSTIFYING THE REACT LOOP

The ReAct loop (Yao et al., 2022) is a structured sequence of text-based operations. There is no hidden "magic"—it is a simple "Text In → Text Out" cycle that allows for sophisticated, autonomous behavior.

STEP	NATURE	FUNCTION
THOUGHT	NATURAL LANGUAGE	The model generates tokens to reason about the goal and plan the next logical step.
ACTION	STRUCTURED JSON	The model outputs a specific request for a tool call (e.g., search_database).
OBSERVATION	TOOL RESULT	The system executes the tool and feeds the real-world data back into the model's context.

REPEAT UNTIL: "I HAVE ALL THE INFORMATION. HERE IS THE FINAL ANSWER."

CLAUDE CODE'S MINIMALIST TOOLSET

Claude Code is powered by a compact set of developer-like tools. With terminal access, the LLM can implement features and diagnose issues autonomously.

TOOL NAME	PURPOSE & CAPABILITY
read_file(path)	Accesses file contents for analysis. Returns the text within a specific file to the model's context.
write_file(path, content)	Modifies or creates project files. The primary mechanism for the agent to implement code changes.
run_command(cmd)	Executes bash commands. Allows the agent to run tests, check diffs, and compile code.
search_code(query)	Navigates large codebases via grep or ripgrep. Essential for finding relevant files.